

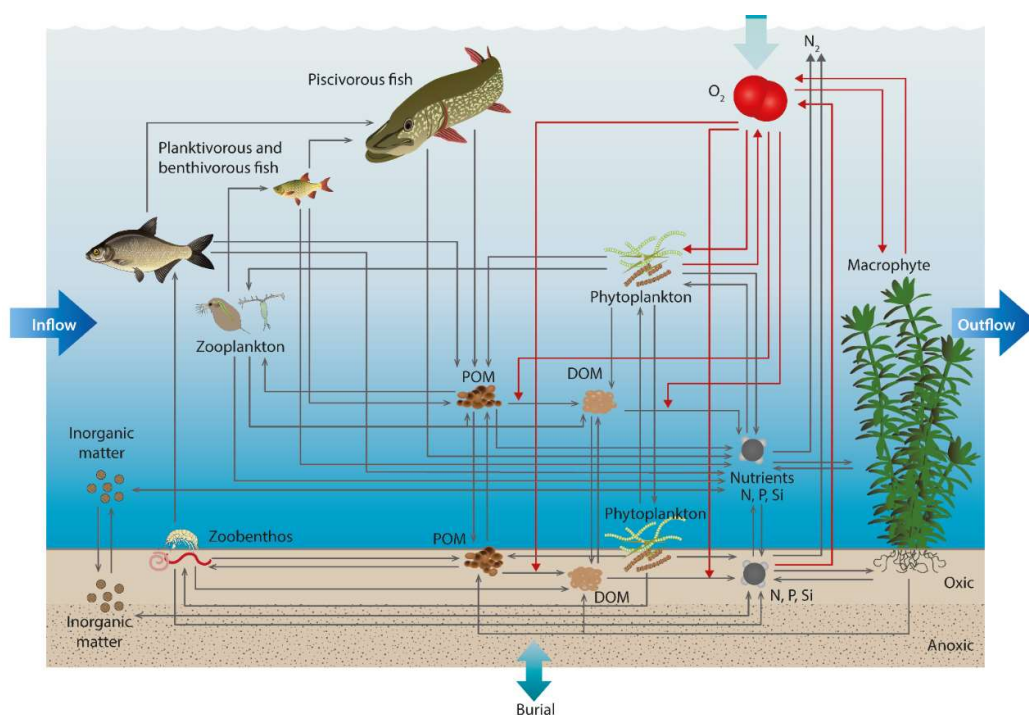
Program for 3-day workshop on:

Modelling lake ecosystems with WET and PCLake

Purpose of the workshop

This workshop trains participants in the theory and practical application of the state-of-the-art lake ecosystem model Water Ecosystems Tool (WET). The workshop covers the theory behind as well as model development of the lake models PCLake and WET. Through combined lectures and hands-on exercises on participants own computers, participants will learn how to

- Set-up a WET lake model via the QGIS interface QWET
- Manual calibration and scenario executions with WET models
- Auto-calibration and sensitivity analysis of WET models with parsac
- Leverage the modularization of WET and set-up different food webs
- Contribute to the development of WET modules



The software used has been developed for Windows computers, and participants therefore need a Windows computer to be able to follow the hands-on exercise.

Workshop instructors

Your workshop instructors will be Postdoctoral researcher Dr Tobias K Andersen (tkan@aquaa.dtu.dk) and Senior researcher Dr Xiangzhen Kong (xzkong@niglas.ac.cn). They both have great experience with model development and application for lakes and

reservoirs all over the world.

Dates

12th to 14th April 2025

Workshop location

Institute for Ecological Research and Pollution Control of Plateau Lakes
Yunnan University, Chenggong, Kunming

The workshop is free and open for students and researchers with basic knowledge of aquatic ecosystems and interest in ecological modelling of waterbodies.

Tentative agenda for the workshop

Datetime	Content
Day 1	
9:00-9:15	Introduction to the workshop and instructors
9:15-10:00	Lecture 1: Theory and use of PCLake
10:00-10:30	Break and group photo
10:30-12:00	Lecture 2: Theory on lake ecosystem modelling with focus on WET
12:00-13:30	Break
13:30-14:00	Tutorial 1: QWET setup
14:00-16:00	Hands-on exercise 1: Basic set-up of WET with QGIS and the Shahe example
16:00-17.30	Prepare for day 2: Install parsac, pp and SALib
Day 2	
9:00-9:15	Catch-up on the previous day
9:15-10:15	Lecture 3A: Calibrating lake ecosystem models: Recommendations for WET users
10:15-10:30	Break
10:30-11:00	Tutorial 2: Introduction to parsac and auto-calibration
11:00-12:00	Hands-on exercise 2: Auto-calibration of your WET model
12:00-13:30	Break
13:30-14:15	Lecture 3B: Evaluating lake ecosystem models: Recommendation for WET users
14.15-14.45	Break
14.45-16.00	Continue and finish exercise 2
16.00-17.30	Q&A
Day 3	
9:00-9:15	Catch-up on the previous day
9:15-10:00	Tutorial 3: Introduction to parsac and sensitivity analysis (SA)
10:00-10:15	Break
10:15-11:15	Hands-on exercise: Perform a SA of your WET model
11:15-12:00	Lecture 4: Why all modelers should do sensitivity analysis
12:00-13:30	Break
13:30-14:15	Lecture 5: Adapting WET to your case study: WET modules and simulating restoration measures
14:15-14.30	Break
14:30-15:00	Tutorial 4: Food web configurations and state variable manipulations in

	WET
15:00-15:30	Lecture 6: Outlook and future of WET
15:30-16:00	Closing remarks of the workshop
16:00-17:30	Q&A

Workshop materials

To participate in the hands-on part of the workshop of day 1, you will need to have a working version of QGIS (version 3.34) and the graphical user interface QWET. **In the morning on day 1, we will provide QGIS and QWET software for you to install.** If you are a Mac user, we highly recommend you apply a windows-partition on your system, as the QWET installer is only available in windows format.

On day 1, we provide video tutorials on configuring, calibrating and running scenarios of GOTM-WET models with QWET. We therefore recommend bringing a headset, so you can work on the exercises while viewing the video tutorials by yourself.

Below you can find a description of the software required for day 1.

QGIS

To run QWET you need to install QGIS3 64-bit version 3.16 or newer (available here: www.qgis.org). We recommend you install QGIS3 64-bit version 3.34 (QWET is currently tested up to version 3.36).

QWET

QWET is an open source QGIS plugin for application and user adaptation of the Water Ecosystems Tool (WET). The QWET installer is available on the workshop Github page under *Materials*. You will need to have installed QGIS before installing QWET. To enable QWET in QGIS (after QWET has been installed), select Plugins in the top menu in QGIS and select "Manage and Install Plugins". Then, a pop-up window will appear, find QWET and tick of the box next to QWET. Now a QWET logo will appear on the QGIS panel. Enabling QWET is also demonstrated in the video tutorials.

GOTM-WET executables

GOTM-WET executables are available to users that installed QWET. For users not interested in using QWET, GOTM-WET v2.1 executable for 64-bit Windows can be downloaded on the [WET Zenodo repository](#).

Case study data for Shahe Reservoir

For our case study, we will work with the Chinese Shahe reservoir. You can access Shahe data folder in *Materials* folder on the [WET workshop page](#) (under *Shahe_data*). The folder includes files to configure a GOTM-WET model to Shahe Reservoir with QWET with lake specific hypsography and inflow and weather forcing as well as in-lake observations and files with calibrated parameters.

We also recommend also installing PyNCView for the first day.

PyNCView

To visualize model output with contour plots, we recommend using the software PyNCView. You can download an installer (for Windows) at this page ([pyncview-0.99.9.msi](https://github.com/BoldingBruggeman/pyncview/releases)): <https://github.com/BoldingBruggeman/pyncview/releases> (under Assets). You can also download PyNCView in Python via “pip install pyncview”.

For day 2 and 3, you will need to have installed a working environment of Python as well as the Python packages parsac and parallepython (pp). We will provide USB-sticks with a suitable version of the Python distribution Anaconda, Python and the packages parsac and pp.

If you are interested in downloading the necessary software yourself, you can find a description of the softwares and how to install it below.

parsac

parsac is a Python-based tool for sensitivity analysis and auto-calibration in parallel developed by Bolding&Bruggeman. It is designed for analysis of models that take significant time to run.

parsac can be installed in a working Python environment. We recommend using Python3 via [Anaconda distribution](#). To install the latest released version of parsac, type in your Anaconda prompt window “pip install parsac –user”. Remove “—user” to install in the system's shared Python directory (not recommended). Some systems have multiple versions of pip, e.g., pip for Python 2, pip3 for Python 3. Make sure you use the command that corresponds to the Python version you want to install into.

For day 2 and 3, you need to install Python 3.12 to use parsac.

We recommend using Python3 via [Anaconda distribution](#):

1. Click here [Anaconda distribution](#)
2. Optional if you will register for Anaconda (else click “skip registration”)
3. Download and install Anaconda with Python 3.12

Once Anaconda is installed, open an Anaconda prompt window and type “pip install parsac –user”. Python will now download and install parsac (you need internet connection for this).

Remove “—user” to install in the system's shared Python directory (not recommended). Some systems have multiple versions of pip, e.g., pip for Python 2, pip3 for Python 3. Make sure you use the command that corresponds to the Python version you want to install into.

parsac dependencies

parsac supports parallel simulations through [Parallel Python](#). This package supports Python 2 out of the box (pip install pp --user), but its Python 3 version is currently in beta. To install pp in Python 3, either [download the zip file with the Python 3 port of Parallel Python](#) or use the pp version provided on the USB stick today (“pp-1.6.4.4.zip”):

1. Extract contents of pp-1.6.4.4.zip,
2. Open a command prompt there and navigate the folder with extracted pp-1.6.4.4 directory,
3. Execute command run "python setup.py install".

parsac uses [SALib](#) for sensitivity analysis. Typically, this can be installed with pip install SALib --user. If you are using [the Anaconda Python distribution](#), you can instead do "conda install SALib" (you may need to add "-c conda-forge").

For more details on parsac and its installation, we recommend checking out its Github page: <https://github.com/BoldingBruggeman/parsac>.

If you have problems installing the required software or downloading the case study, you are welcome to contact us via the [Github Issues board](#).

More resources

Water Ecosystems Tool (WET)

WET is a further development of FABM-PCLake by Hu et al., at Aarhus University, Denmark. Key features originate from the PCLake aquatic ecosystem model by Janse and van Liere (1995), but key features and inspiration from CAEDYM and Ecopath/Ecosim has also been implemented. WET can describe interactions between multiple trophic levels and abiotic nutrient dynamics in both the water column and the sediment. The model accounts for the dynamics of dry weight, nitrogen, phosphorous, silica and oxygen, and features bottom-shear-dependent resuspension, as well as two different light-limitation functions for phytoplankton. WET is also implemented within the FABM framework, allowing the model to be coupled to various physical driver models, e.g. GOTM (1D, Burchard et al., 1999) or GETM (3D, e.g. Stips et al., 2004), without changing any of the model code. For introduction to WET, we refer you to the scientific articles:

Schnedler-Meyer, N. A., Andersen, T. K., Hu, F. R. S., Bolding, K., Nielsen, A., & Trolle, D. (2022) *Water Ecosystems Tool (WET) 1.0 – a new generation of flexible aquatic ecosystem model*. [Geoscientific Model Development](#)

Schnedler-Meyer, N. A., & Andersen, T. K. (2024) *Dining in danger: Resolving adaptive fish behavior increases realism of modeled ecosystem dynamics*. [Ecology and Evolution](#)

General Ocean Turbulence Model (GOTM)

Most commonly WET is coupled to the lake branch of 1D hydrodynamic model GOTM (source code available at <https://gitlab.com/watertech-public/waterecosystemstool/gotm>). For details on lake branch GOTM, see the models Gitlab page. For more information on GOTM in general, check out <https://gotm.net/portfolio/>.

For a brief description of GOTM and how it compares to other 1D hydrodynamic models, we recommend

Feldbauer, J., Mesman, J.P., Andersen, T.K., Ladwig, R. (2025) [Learning from a large-scale calibration effort of multiple lake models](#). Hydrol. Earth Syst. Sci.

Framework of Aquatic Biogeochemical Models (FABM)

The Framework for Aquatic Biogeochemical Models (FABM, developed by Bolding & Bruggeman) allows coupling a biogeochemical model to a wide variety of hydrodynamic models in 0D, 1D, 2D or 3D, without changing any model code, and encourages and supports modularization of ecosystem models. For more details, we refer you to FABMS wiki page <https://github.com/fabm-model/fabm/wiki>.

Model study on the Shahe Reservoir

You can find more information on the model study of Shahe Reservoir, which this workshop is based around, in the paper

Cui, Y., Zhu, G., Li, H., Luo, L., Cheng, X., Jin, Y., & Trolle, D. (2016). Modeling the response of phytoplankton to reduced external nutrient load in a subtropical Chinese reservoir using DYRESM-CAEDYM. [Lake and Reservoir Management](#), 32(2), 146–157.